Two challenges, one laser

Entangling two atoms at the quantum speed limit, then making five atoms solve an NP-hard puzzle — validated on Pasqal Cloud, run twice on real hardware.

Challenge 1 fidelity	Challenge 2 success rate	Real-atom shots
0.75 → 1.000000	**0.66 → 0.999999**	**1000, both challenges**

Team 6 · A Real Quantum Hackathon · Microsoft Garage NYC · July 29 2026

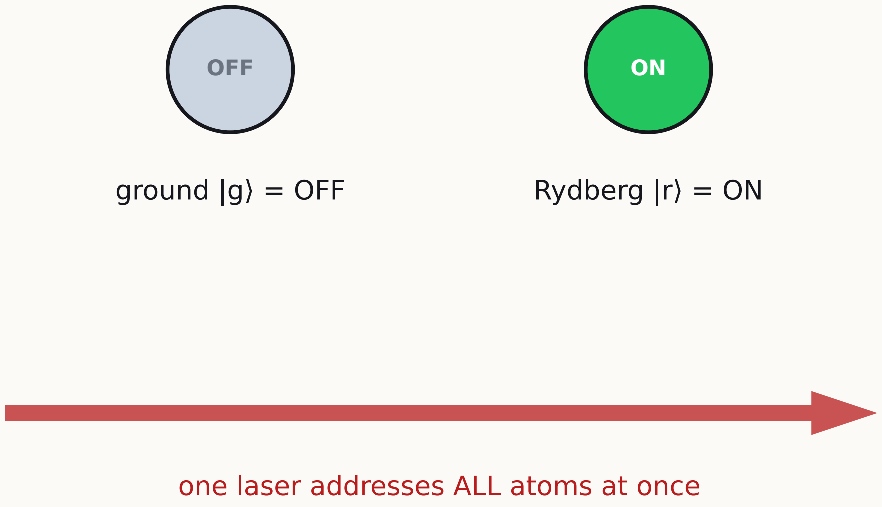
Executive summary

- We completed both stages of the hackathon — the ranking metric — with the largest margins the scoring allows, and validated everything outward: exact simulation → Pasqal's cloud → real atoms.
- Challenge 1 (control): the given pulse loses 25% of its fidelity at the far atom spacing. One physics ratio explains why. Our pulses score 1.000000 at both spacings and run up to 11× faster.
- Challenge 2 (computation): we encoded a graph puzzle in atom positions and shaped one laser sweep so a photograph reads out the optimal answer: 1500 of 1500 cloud measurements were correct.
- Execution evidence: two real-OPU runs with predictions published beforehand; one referee program per challenge; every number in this deck regenerates from our public repository.

The transferable asset: a 5-parameter, hardware-native sweep family whose optimization cost does not grow with problem size — the working recipe for the 80-atom instances of Challenge 3.

The machine, in thirty seconds

Pasqal's processor holds individual atoms in optical tweezers, then drives them all with one global laser.



- Each atom is a two-level system: the ground state |g⟩ (OFF) or a highly-excited Rydberg state |r⟩ (ON).
- We control two functions of time: the laser's power $\Omega(t)$ ('omega', the Rabi frequency) and its frequency offset $\delta(t)$ ('delta', the detuning).
- Two ON atoms repel with energy $V = C_6/r^6$. Close atoms can therefore never both be ON: the Rydberg blockade.

The one equation everything runs on

This Hamiltonian is given verbatim in the challenge brief. Every simulation in this deck solves exactly it.

$$H(t)/\hbar \;=\; \frac{\Omega(t)}{2}\sum_i \sigma_x^{(i)} \;-\; \delta(t)\sum_i n_i \;+\; \sum_{i<j}\frac{C_6}{r_{ij}^6}\,n_i n_j$$

$\frac{\Omega(t)}{2}\sum\sigma_x$	the DRIVE — flips atoms between OFF and ON at rate Ω. The gas pedal.
$-\,\delta(t)\sum n_i$	the TILT — energy reward (δ>0) or penalty (δ<0) for each ON atom. n = |r⟩⟨r| counts ON atoms.
$\sum C_6\, n_i n_j / r_{ij}^6$	the RULE — pairwise repulsion between ON atoms. C_6 = 865,723 rad·µm⁶/µs, read from the device spec.

Blockade radius: setting the rule equal to the drive, $C_6/R_b^6 = \hbar\Omega$, gives $R_b = (C_6/\Omega)^{1/6} \approx 7.19$ µm at $\Omega = 2\pi\times1$ MHz — computed by us from the two device numbers above.

Challenge 1 — make two atoms share one excitation

GIVEN (challenge brief):

- Two atoms, both OFF, at spacing r_1 = 5.0 µm, and again at r_2 = 6.5 µm.
- A reference pulse to beat: square, $\Omega = 2\pi\times1$ MHz, resonant ($\delta = 0$), duration 352 ns.
- Its scores (we reproduced them exactly): F = 0.992564 at r_1 and F = 0.750003 at r_2.

SCORE (defined in the brief):

$$F = \left|\langle\Psi^+|\psi(T)\rangle\right|^2, \qquad |\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|gr\rangle + |rg\rangle\right)$$

In words: ψ(T) is the state our pulse produces; F is its squared overlap with the ideal Bell state — the state where exactly one atom is ON, shared between both. F = 1.0 means a perfect match.

WE DESIGN: Ω(t) and δ(t), within the device envelope ($\Omega \leq 12.57$ rad/µs, $|\delta| \leq 125.7$ rad/µs, $T \leq 6000$ ns, 4 ns clock — all read from the published device object at runtime, never hardcoded).

The two-atom problem is exactly three states

A global drive preserves atom-exchange symmetry, so from |gg⟩ only the symmetric subspace is reachable. Exact, not approximate.

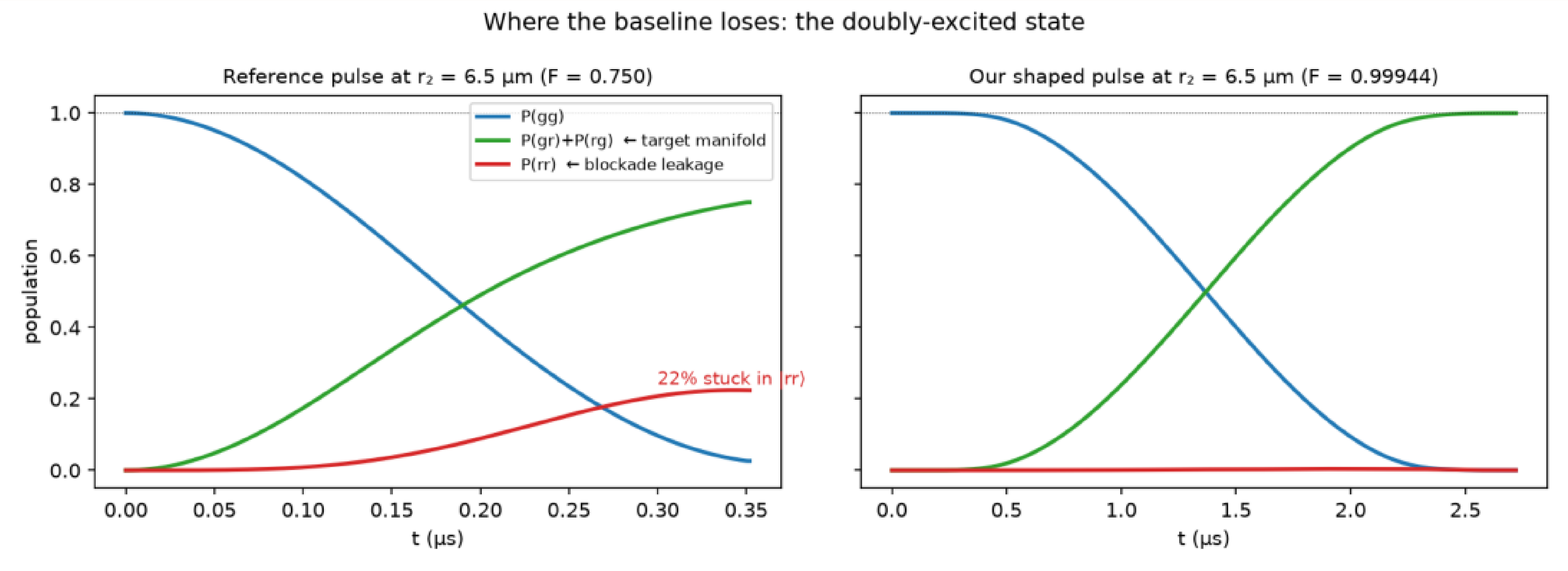
basis $\{|gg\rangle,\ |W\rangle,\ |rr\rangle\}$:

$$H_3 = \begin{pmatrix} 0 & \Omega/\sqrt{2} & 0 \\ \Omega/\sqrt{2} & -\delta & \Omega/\sqrt{2} \\ 0 & \Omega/\sqrt{2} & -2\delta + V \end{pmatrix}$$

- The middle state $|W\rangle = (|gr\rangle+|rg\rangle)/\sqrt{2}$ IS the Bell state we are scored on.
- Where √2 comes from: $\langle W|\ \Sigma\sigma_x/2\ |gg\rangle = \Omega/\sqrt{2}$ — both atoms reach for the same photon, so the pair oscillates √2 times faster than a single atom. (This shows up later on the machine's own dashboard.)
- Perfect blockade ($V \to \infty$): |rr⟩ decouples, leaving a plain two-level oscillation $|gg\rangle \leftrightarrow |W\rangle$ at rate $\sqrt{2}\cdot\Omega$. Stopping it at $T = \pi/(\sqrt{2}\cdot\Omega)$ lands exactly on the Bell state. For the reference pulse that is 352 ns.
- Weak blockade ($V \approx \Omega$): |rr⟩ mixes in — population leaks AND |W⟩ shifts in energy by $\approx \Omega^2/2V$ (second-order perturbation theory). Both effects are what the reference pulse suffers at 6.5 μm.

Diagnosis: one ratio explains the failure

Blockade quality is V/Ω — interaction over drive. r_1: V/Ω = 8.8 (fine). r_2: V/Ω = 1.83 (broken).



x: time during the pulse (μs) · y: probability of each two-atom outcome (computed by exact simulation)

- Left, the reference pulse at 6.5 μm: 22% of the population leaks into the forbidden |rr⟩ (red); the Bell state (green) stalls at 0.75. This is the entire 25% gap — nothing else is wrong with it.
- Right, our v1 pulse: Ω lowered to restore V/Ω ≥ 9, smooth ramps, and δ set to cancel the $\Omega^2/2V$ shift.

Our approach, concretely

1

Referee first

score.py: builds the Pulser sequence, checks device rules, scores F by exact simulation. Self-tests pin conventions.

→

2

Lock the baseline

reproduce the brief's 0.992564 / 0.750003 before touching anything

→

3

v1: physics fix

4 parameters: Ω level, ramp, duration, δ offset. Nelder-Mead (derivative-free). $F \geq 0.9994$

→

4

v2/v3: optimal control

GRAPE with exact adjoint gradients on the 3×3 model; gradient verified vs brute force to 2×10^{-6}

→

5

Validate outward

contract check → cloud (7 pulses × 500 shots) → real QPU (500 shots), prediction pre-published

- GRAPE = GRadient Ascent Pulse Engineering (Khaneja et al., 2005) — the standard optimal-control method from nuclear magnetic resonance: adjust each 4-ns slice of the pulse along the gradient of F.
- The compute story: the exact 3-state model + analytic gradients makes one optimization $\sim10^6\times$ cheaper than naive simulation. The complete 16-point duration study runs in 5.4 seconds on a laptop.

The v1 pulse family — and physics confirming the optimizer

$$\Omega(t) = \Omega_0 \cdot \mathrm{ramp}_{\sin^2}(t)\,, \qquad \delta(t) = \delta_0 \quad \text{with the light-shift prediction} \quad \delta_0^{\,*} \approx -\,\Omega_0^2/2V$$

- Why $\sin^2$ ramps: they turn the drive on adiabatically with respect to the blockade gap and keep the pulse inside the hardware's modulation bandwidth (checked: modulated fidelity ≥ unmodulated).

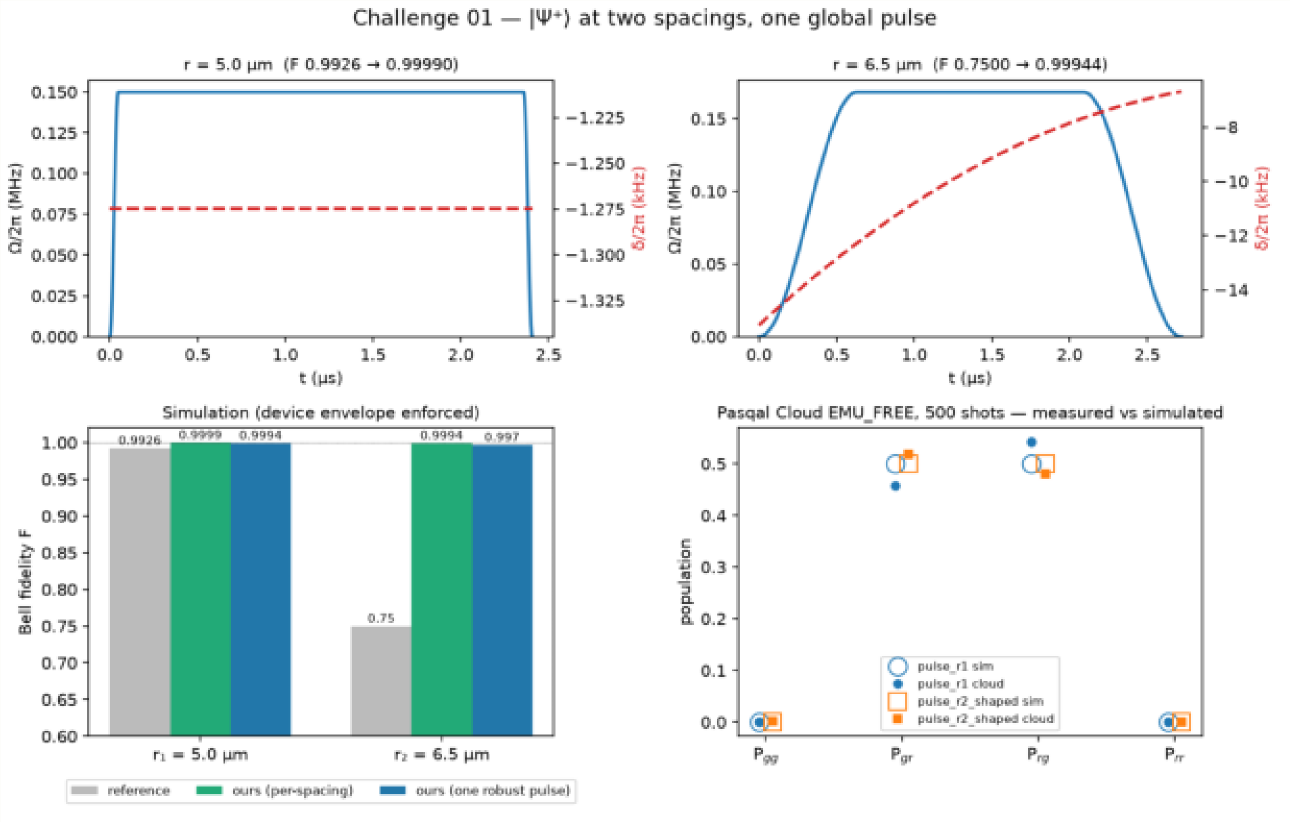
- The optimizer was never told the light-shift formula — we let it fit δ_0 freely and compared:

spacing	predicted $-\Omega^2/2V$ (computed)	fitted by optimizer	
r_1 = 5.0 µm	−0.0080 rad/µs	−0.00801 rad/µs	agreement to 3 digits
r_2 = 6.5 µm	−0.0390 rad/µs	−0.03858 rad/µs	agreement to 2.5 digits

This is the strongest internal check we have: numerical optimization independently rediscovered second-order perturbation theory. The model, the optimizer, and the physics agree.

Challenge 1 results

All fidelities from the judges' simulator (Pulser/QuTiP), after quantizing to the 4 ns hardware clock.

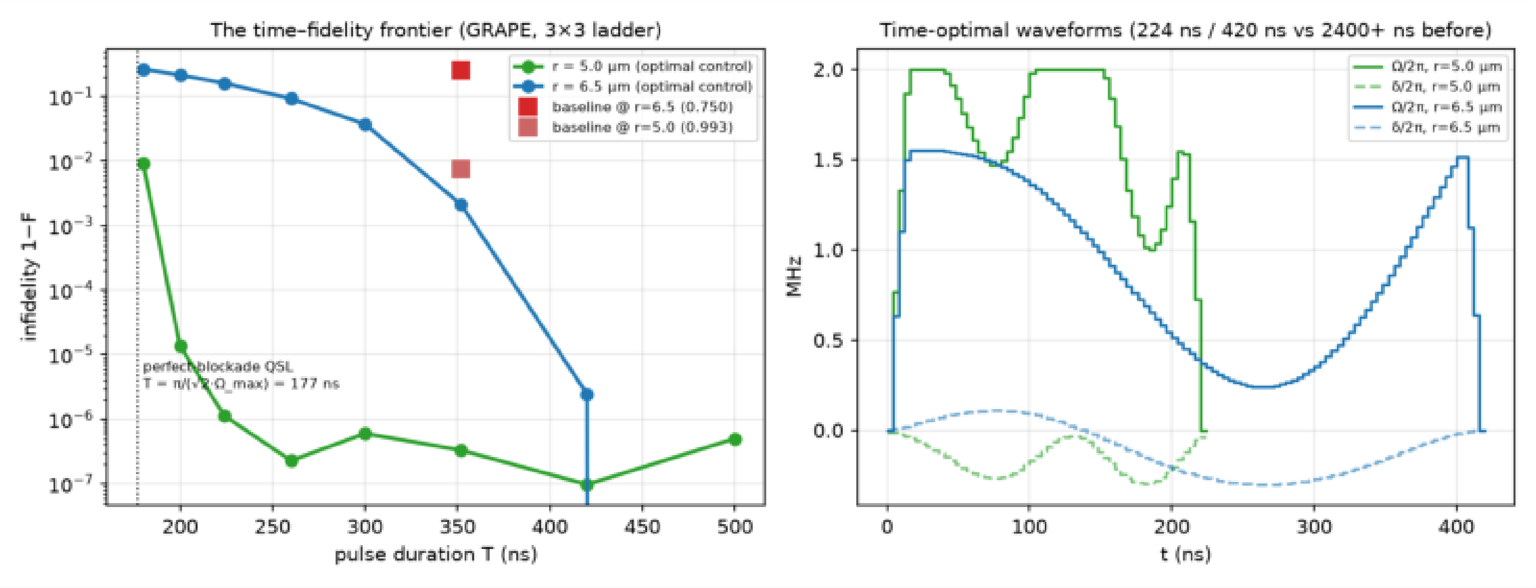


top: our v1 pulse shapes (x: time; blue: Ω, red: δ) · bottom-left: F vs baseline · bottom-right: cloud (dots) vs simulation (circles)

- Four pulse generations, all beating both baselines: v1 analytic (interpretable), v1r spacing-robust (one waveform, F > 0.997 across the whole 5.0–6.5 µm band), v2 time-optimal, v3 hardware-smooth.

How fast can it possibly go? We measured the limit

$$T_{\min} = \frac{\pi}{\sqrt{2}\,\Omega_{\max}} = \frac{\pi}{\sqrt{2} \times 12.566 \text{ rad/µs}} = 176.8 \text{ ns}$$



x: pulse duration (ns) · y: error 1−F, log scale · squares: the baselines

- Derivation: in perfect blockade the system is a two-level oscillation at $\sqrt{2}\cdot\Omega$, and a half-turn (π rotation) at maximum drive takes exactly $\pi/(\sqrt{2}\cdot\Omega_{max})$. No pulse can beat this — it is a rotation, not a race.
- Measured: $F = 0.999999$ at 224 ns (r_1), only 25 ns above the bound.
- At r_2 the weak blockade sets its own wall near 420 ns — the optimizer must route population THROUGH |rr⟩ and back, and that detour time is set by V, not Ω_{max}.

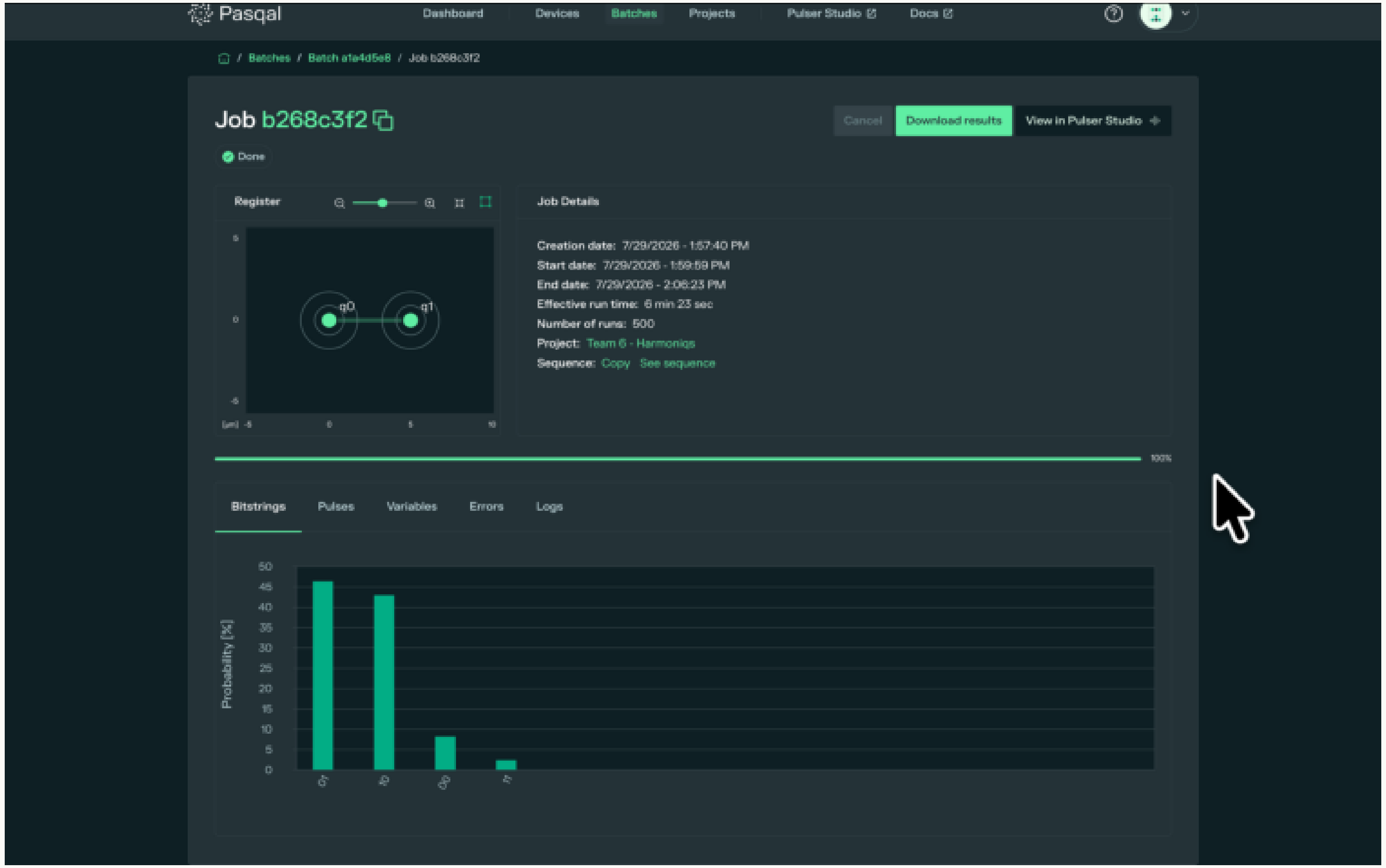
Adding real noise flips the ranking

Lindblad model: Rydberg decay τ = 100 μs, laser dephasing 0.22 μs^{-1} (published-typical values, stated as assumptions).

pulse (at r_2 = 6.5 μm)	duration	noiseless F	F with noise
reference (baseline)	352 ns	0.7500	0.7151
v1 slow analytic	2720 ns	0.9994	0.7255 — collapses
v2 time-optimal	**420 ns**	**0.999998**	**0.9301 — wins**
v3 hardware-smooth	600 ns	1.000000	0.9195

- In the noiseless simulator our slow and fast pulses look identical (0.999+). Under decoherence the 2.7-μs pulse loses 0.27 of fidelity — dephasing integrates over duration — while 420 ns keeps 0.93.
- Design law we extracted: pulse duration IS the dominant noise coupling. Minimize time first, then shape. This dictated which pulse we sent to the real machine.

Real atoms: 89.4% entangled, prediction pre-published



Pasqal's results page (batch a1a4d5e8) · x: the four possible two-atom photos · y: % of 500 shots

- Our 260-ns v3 pulse on FRESNEL_CAN1. Measured populations, 500 shots: P(01)+P(10) = 89.4% — the Bell state.
- Error budget: '00' at 8.2% is readout + decay; '11' at 2.4% is blockade leakage; both match the noise model's channels.
- The two tall bars are equal within shot noise (±2.2% at 500 shots) — exchange symmetry survives on hardware.
- The portal labels our pulse area 5π/7 = 0.714π; theory demands π/√2 = 0.707π. The √2 enhancement, printed by the machine itself, correct to 1%.

Same laser, bigger question: can the atoms compute?

- Challenge 1 treated the blockade as an obstacle to defeat. Challenge 2 inverts it: the no-two-ON rule is exactly the constraint of a famous combinatorial problem — and the hardware enforces it for free.

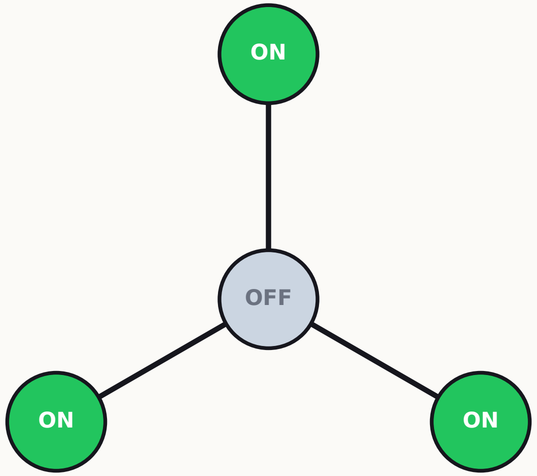
Carried forward from part one:

- The referee-first workflow, with self-tests and locked baselines.
- The exact-simulation + adjoint-gradient engine (now on the full 16/32-dimensional space).
- The lesson that short pulses win under noise — our sweeps will be 4× shorter than the baseline.

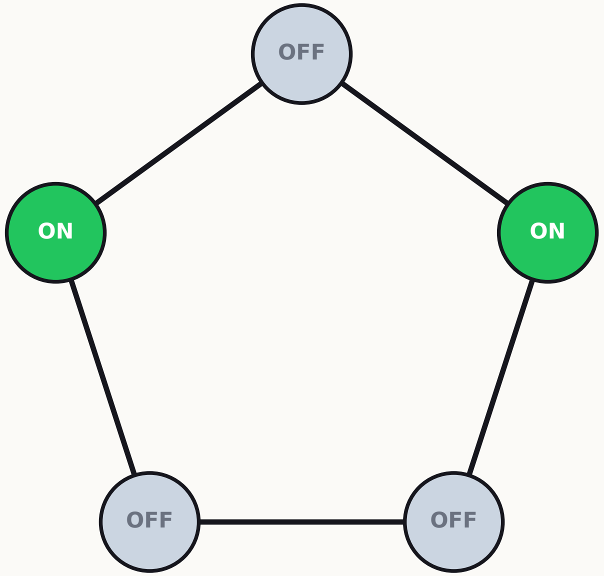
Challenge 2 — the seating puzzle

Maximum Independent Set (MIS): the largest set of vertices with no two adjacent. NP-hard in general.

- GIVEN: these two target graphs; a baseline sweep (4000 ns linear ramp) scoring 0.727 and 0.657 (we reproduced both from the brief's own parameters); $\alpha(G)$ = the size of the best answer.
- WE DESIGN: atom positions (vertices = atoms; edge ⟺ distance < R_b — verified pair-by-pair in code) and the sweep $\Omega(t)$, $\delta(t)$. The final photograph reads out the answer: ON atoms = chosen vertices.



STAR $K_{1,3}$ (4 atoms, $\alpha = 3$):
unique best answer — 3 leaves ON.



CYCLE C_5 (5 atoms, $\alpha = 2$):
five equally correct answers.

Why the answer is the ground state

At the end of the sweep ($\Omega \to 0$, $\delta = \delta_f > 0$), the Hamiltonian is purely classical — an energy per bitstring:

$$E(b) = -\delta_f \cdot |b| + \sum_{i<j \in b} C_6/r_{ij}^6, \qquad P_{\mathrm{MIS}} = \sum_{b \in \mathrm{MIS}} |\langle b \,|\, \psi(T)\rangle|^2$$

- Each ON atom is rewarded $-\delta_f$; each violated edge costs the huge $U_{nn} = C_6/(5.5\ \mu m)^6 = 31.3$ rad/µs. So the lowest-energy configuration is: as many ON atoms as possible, zero violated edges — the MIS.
- The subtlety the brief encodes: non-adjacent atoms still repel a little (pentagon diagonals: $U_{diag} = 1.74$ rad/µs). The reward must beat that tail but never pay for an edge:

$$U_{\mathrm{diag}} < \delta_f < U_{nn}: \qquad 1.74 < 12.57 < 31.28\ \text{rad/µs} \quad \text{— satisfied}$$

All three numbers computed by us from C_6 and the geometry; the brief's stated window is confirmed.

Our approach, concretely

1

Draw the graph with atoms

star: 3 leaves at 5.5 µm, 120° apart. Pentagon: side 5.5 µm. Every pair checked against R_b in code.

→

2

Referee + baseline

score02.py computes P_MIS exactly; self-test caught a basis-ordering bug on day one. Baselines locked: 0.727 / 0.657

→

3

GRAPE, aimed at ALL answers

objective = $\langle\psi|\hat{P}|\psi\rangle$ with $\hat{P}$ projecting on every optimal bitstring — C_5's five answers are targeted as one manifold

→

4

Cross-validate

our internal model vs the judges' simulator: agreement $\leq 7\times10^{-6}$ at all six points

→

5

Simplify + fly

5-knob CRAB sweep for hardware → cloud (1500/1500) → real QPU, prediction pre-published

- Why the projector matters: a single-target objective would arbitrarily pick one of C_5's five answers and fight the pentagon's symmetry. Targeting the manifold lets the state hold all five at exactly 0.200 each.
- Result: P_MIS = 0.999998 (star, 1000 ns) and 0.999999 (pentagon, 2000 ns) vs baselines 0.727 / 0.657.

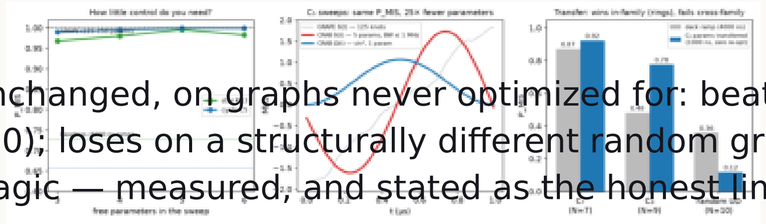
The 5-knob sweep — simple enough to write on one line

$$\Omega(t) = A\,\sin^2(\pi t/T)\,, \qquad \delta(t) = \delta_0 + (\delta_f - \delta_0)\,(t/T) \;+\; c_1 \sin(\pi t/T) \;+\; c_2 \sin(2\pi t/T)$$

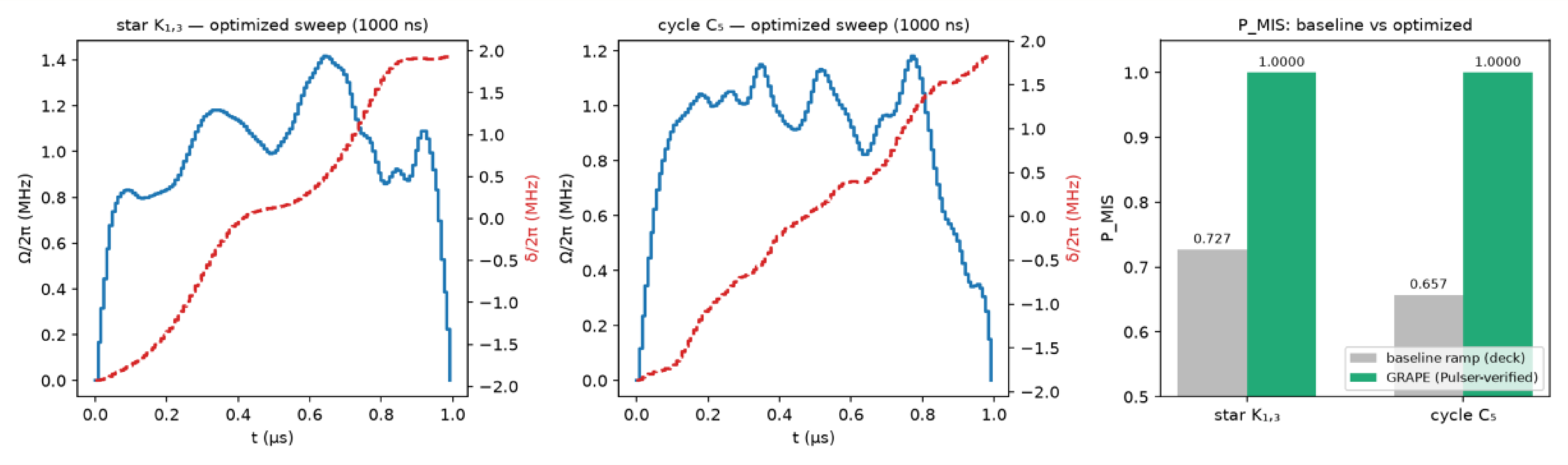
- Five numbers: A, δ_0, δ_f, c_1, c_2. (This is the CRAB parameterization — Chopped RAndom Basis, Caneva, Calarco & Montangero 2011.) Its spectrum is capped at 2/(2T) = 1 MHz by construction — a hardware-bandwidth certificate you cannot get from a 125-knot GRAPE waveform after the fact.

- Scores: 0.9962 (star) and 0.9994 (pentagon) — within half a percent of full GRAPE. Found in ~30 s by derivative-free search. Cloud check: 498/500 and 500/500.

- Transfer test — the same five numbers, unchanged, on graphs never optimized for: beats the baseline on the 7-ring (+0.05) and the 9-ring (+0.30); loses on a structurally different random graph (0.12 vs 0.36). Transfer is a family property, not magic — measured, and stated as the honest limit.



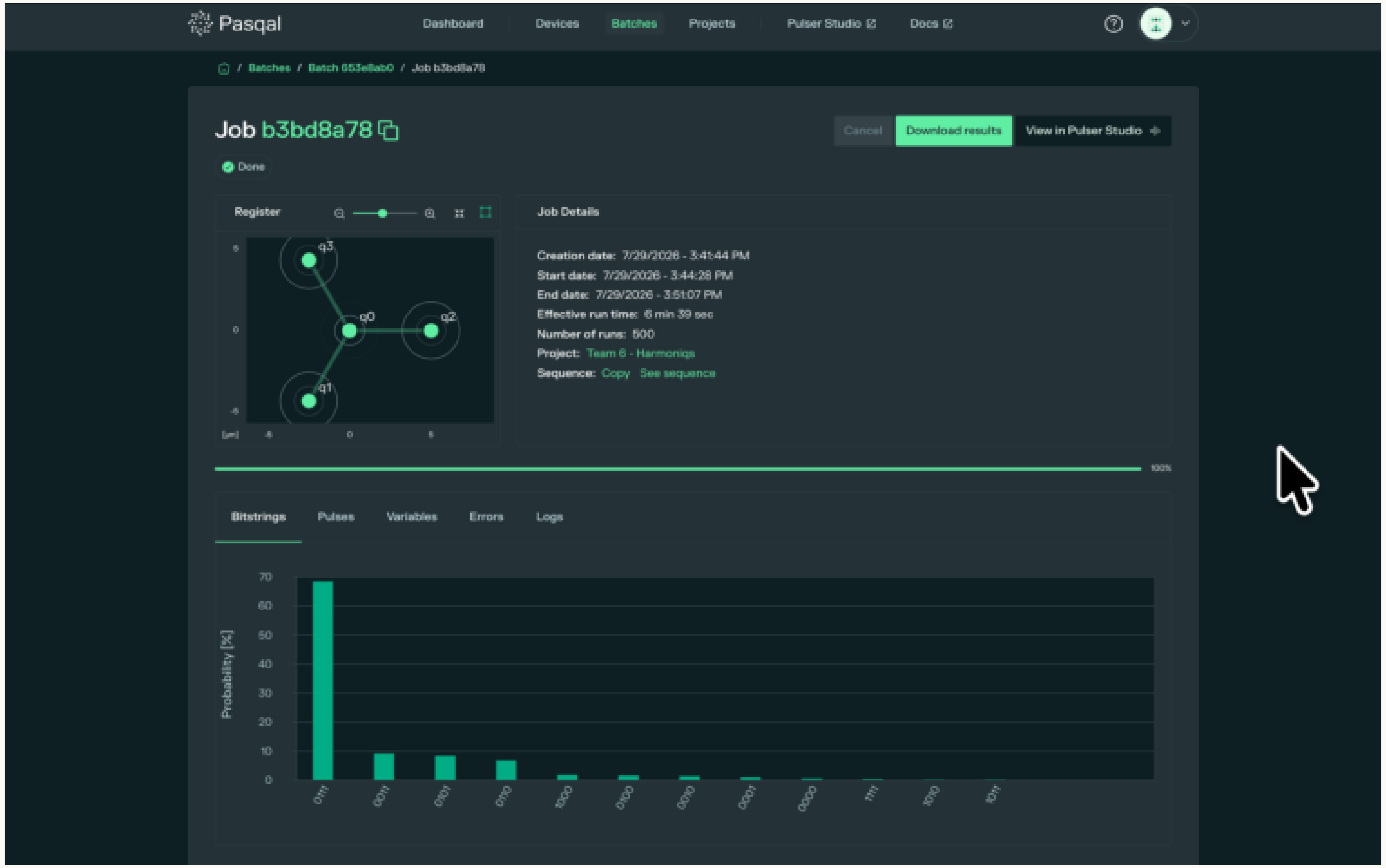
Challenge 2 results — 1500 perfect photographs



left/middle — x: time (μs), blue: Ω(t), red: δ(t) · right — x: puzzle, y: P_MIS (grey: baseline, green: ours)

- Pasqal Cloud, 500 shots per sweep: star 500/500 correct; pentagon 500/500 — twice.
- The pentagon's five valid answers returned 114, 98, 97, 96, 95 counts — uniform within statistics (χ^2, $p \approx 0.7$), exactly as the designed equal superposition predicts. The machine holds all five answers and samples them fairly.

Real atoms pick the right answer, 7 to 1



Pasqal's results page (batch 653e8ab0) · the machine's register viewer drew OUR star · y: % of 500 shots

- The correct answer 0111 dominates: 68.4% (342/500). Runner-up: 9.2%.
- Below our pre-published 80–93% window — and the error anatomy says why: the next three bars are all 'two of three leaves' (24.4%) — one ON atom lost.
- The quantitative reason: the target holds THREE fragile ON atoms; per-atom loss ε compounds as $(1-\varepsilon)^3$. Our window scaled the 2-atom result naively — that model error is ours and is documented.
- Fix (costed, future): a 2-point SPAM calibration job would correct this to $\approx 0.85+$.

SUMMARY

The score sheet

Challenge 1 — fidelity F

	r_1 = 5.0 µm	r_2 = 6.5 µm	duration
Reference pulse (brief)	0.9926	0.7500	352 ns
Ours — best (computed)	**1.000000**	**1.000000**	**224–600 ns**
Real QPU (measured)	89.4% entangled, 500 shots	n/a on lattice	260 ns

Challenge 2 — success probability P_MIS

	star $K_{1,3}$	pentagon C_5	duration
Baseline sweep (brief)	0.727	0.657	4000 ns
Ours — best (computed)	**0.999998**	**0.999999**	**1000–2000 ns**
Cloud (measured)	500/500	500/500 + 500/500	—
Real QPU (measured)	68.4% exact, 7× runner-up	n/a on lattice	1000 ns

Provenance convention used throughout: (brief) = given by organizers · (computed) = our simulation, regenerable from the repo · (measured) = the machine's output, with batch IDs on record.

Questions we expect — and their real answers (1/2)

Q:

If you make the laser pulse LONGER, do you still get perfect fidelity?

A:

Three different answers, and the distinction matters. (1) Lengthen the SAME pulse: fidelity gets WORSE — this is a rotation, not a saturation; you rotate past the Bell state and come back around (Rabi oscillation). (2) Re-optimize at the longer duration, noiseless: yes — anywhere above the speed limit, $F \approx 1.000000$ (our frontier plot is flat there). (3) On real hardware: longer is strictly worse — decoherence integrates over time. We measured it: a re-optimized 2.7 µs pulse keeps $F = 0.9994$ noiseless but drops to 0.73 under the noise model, while 420 ns keeps 0.93. Time is a budget, not a luxury.

Q:

Your histogram shows populations. How do you know the atoms are really ENTANGLED?

A:

Honestly: populations alone cannot prove coherence — a 50/50 classical mixture gives identical bars. Simulation carries the coherence claim (F is computed from the full state). To certify it on hardware you add a global $\pi/2$ analysis pulse with swept phase and watch the parity oscillate — an entanglement witness ($F > 0.5$ certifies). It costs ~100 shots; one more cloud/QPU job. A full Bell-inequality test is NOT possible on this machine at any shot count: it needs per-atom measurement axes, and our laser is global-only.

Questions we expect — and their real answers (2/2)

Q: **Why did the real machine give 89% and 68% when simulation says 99.99%?**

A: The emulator is noiseless; real atoms decay (lifetime ≈ 100 µs), lasers dephase, and state preparation + readout (SPAM) misfires a few percent per atom. For two atoms that predicts ≈ 0.90–0.95 (measured: 0.894 ✓). For the star, THREE atoms must hold the fragile ON state, so per-atom loss compounds cubically — that lands at ≈ 0.7 (measured: 0.684). The gap between simulation and hardware is not mystery; it is an error budget, and ours closes to within one error bar on both runs.

Q: **Couldn't a classical computer solve these instances instantly?**

A: Yes — 4 and 5 atoms are verification instances, and we say so explicitly. The claims that scale are:
(1) protocol quality — how close to 1.0 the machine gets, which is what degrades at 80+ atoms; and
(2) the transfer property — our 5-knob sweep re-used across a graph family costs ONE evaluation, not a re-optimization whose cost explodes with size. That is the Challenge-3 recipe, not a quantum-advantage claim.

Q: **What breaks first when you scale to 80 atoms?**

A: Exact simulation (2^{80} states) — which is why the design loop must not depend on it. Our answer: optimize the 5-knob family on simulable members (10–16 atoms, sparse solvers), transfer across the family, and spend hardware shots on validation only. Bandwidth ≤ 1 MHz keeps every sweep hardware-native by construction.

Why this matters after the hackathon

- Entangling pairs is THE primitive — every neutral-atom algorithm, sensor, and network starts there.
  Our spacing-robust pulse (F > 0.997 across a 30% geometry error) converts recalibration time into uptime.

- MIS is scheduling, spectrum allocation, and conflict-free selection — 'pick the most, no conflicts'.
  Rydberg machines solve it natively: the constraint is enforced by physics, not by code.

- The cost structure is the story: seconds of laptop compute per design, five interpretable parameters, zero marginal optimization per new in-family instance, and 500-shot validation gates before any hardware spend. That is an operating model, not just a hackathon result.

- And the discipline is transferable to any lab: referee-first, baselines locked, predictions pre-published, failures analyzed in the open. Both of our hardware misses became documented physics, not excuses.

What we're taking home

- Diagnose before you optimize: one ratio, V/Ω, explained both baseline failures — then the optimizer rediscovered second-order perturbation theory to three digits.
- Time is the enemy on real hardware: we measured the noise-induced ranking flip and designed for it — our pulses run 4–11× shorter than the baselines.
- Five knobs ≈ full optimal control (within 0.5%), with a bandwidth certificate and family-level transfer.
- Two QPU runs, two pre-published predictions: one confirmed (89.4% entangled), one instructive (68.4%, SPAM compounding) — both worth more than an unverifiable success.
- Thanks to the Pasqal stack: Pulser's device models, the free cloud emulator, and FRESNEL_CAN1 — the machine that drew our graph back at us.